

Executive Summary

Autobrain at a glance

Founded in 2019 in Israel as spin-off from leading AI-firm Cortica

Autobrain provides AI SW products for ADAS and autonomous driving

based on self-learning AI and protected by 250+ patents

at drastically lower cost than current market standards



Igal Raichelgauz - Founder & CEO
Ex-CEO Cortica



Karl-Thomas Neumann - Chairman
Ex-CEO Opel, Continental, VW China



Hilla Tavor - CBO
Ex-VP ADAS Business Development
Mobileye

Reimagining Autonomy - Rethinking AI

SW market for automotive perception reaching \$10bn in 2027

dominated by single player Mobileye, OEMs and suppliers in need of viable alternative

Autobrain's self-learning AI drastically outperforms current solutions

at 40% lower cost, 10x less compute, is HW agnostic, modular in its feature set and reliable in edge cases

Mature and proven technology on the way to serial production

full-scale commercial agreements for high-volume automotive business

Accomplished team of AI experts, automotive executives and semiconductor veterans

over 80% AI engineers, including ex-Mobileye, Stellantis, VW, Aptiv, CEVA and NVIDIA

High-growth company with ~\$140m raised to date

seeking to accelerate growth and expand product offerings in response to high market needs and regulatory requirements



Signatures and self learning AI

Traditional (Supervised) Deep Learning Does Not Scale

Autobrain leads the paradigm-shift from supervised to self-learning AI in the mobility sector

Traditional supervised learning

Manual labeling – subject to high costs, bias and errors

Immense compute requirements - unprofitable in mass-market vehicles

Fixed architecture – vulnerability in edge cases

Black box - no visibility or separability of the problem, one size fits all

Self-supervised learning (signature-based)

Self-learning – independent of labeled data, slashing costs & errors

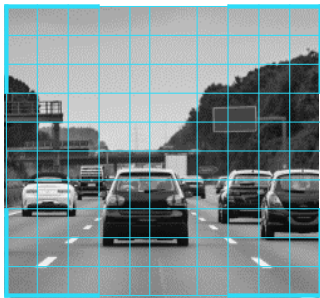
Low compute requirements (up to 10x less) – affordable in any car

Adaptive architecture – full edge case coverage

“Open book” solution – explainable and customizable

Autobrains AI: Self-learning & Signature-based AI

Sensor
Input



Signatures



Concepts
via inter-matching



DB of concept
signatures

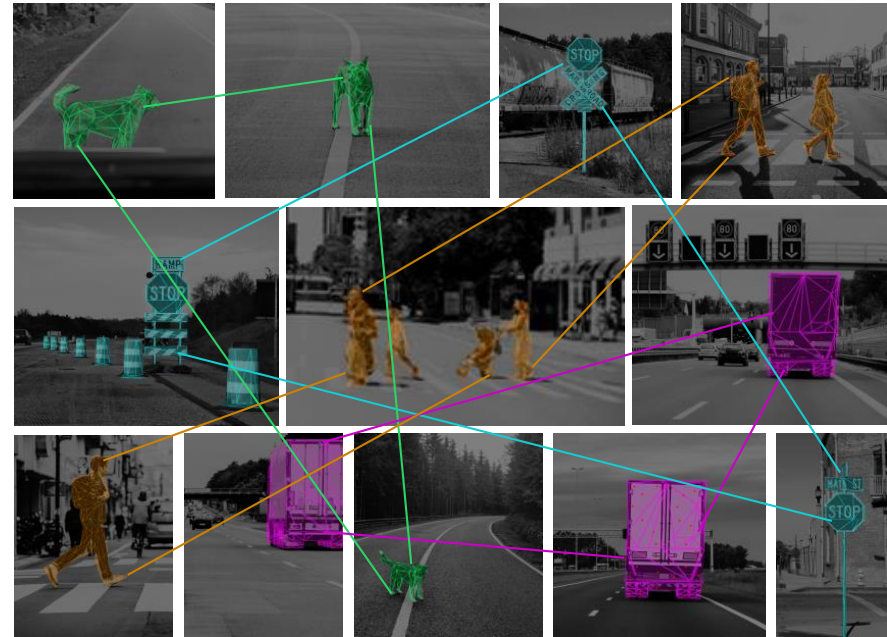


Autobrain AI: Self-learning & Signature-based AI

Mimics human learning by creating generic, lightweight & holistic representations of the world

Bottom-up clustering into signature-based concepts

1. **Data Collection:** Sensory input is independently indexed as compressed neural response, without labeling
2. **Self-Learning:** The system autonomously identifies and matches common patterns inside the neural response
3. **Signatures:** information is saved in low compute, granular representations - not in heavy, generalized semantic labels



- Generic – the same signatures for all tasks
- Scalable – minimal data requirement
- Explainable – the signatures can be investigated and solved

- Compressed – low compute
- Updatable – new objects and can be added without change
- Robust – Comprehensive representation of long-tail edge cases

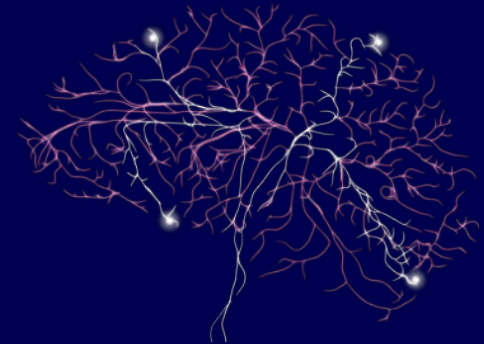
Adaptive Architecture

- The perception architecture is not hand crafted, but self-learned, driven by data
- Different (small fractions of) resources in the network are applied or routed during processing

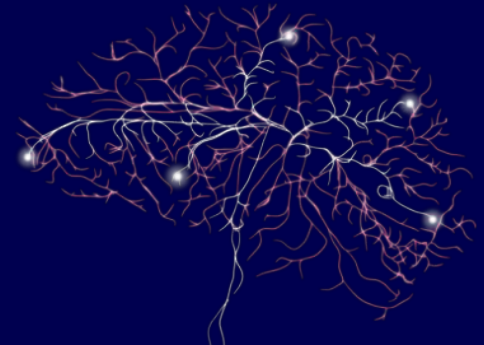
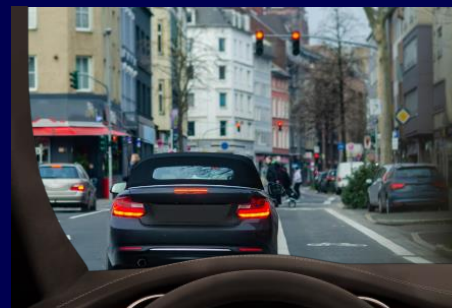
Results in:

- Adaptivity to edge cases
- Efficiency and low compute

Processing Scenario A



Processing Scenario B

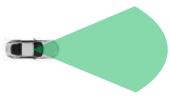
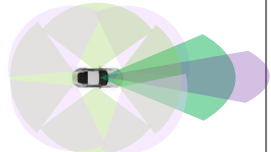
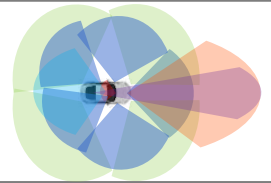


Autobrain's Product Roadmap

Mature and proven technology on the way to serial production
Full-scale commercial agreements for high-volume automotive business

AB SW Products

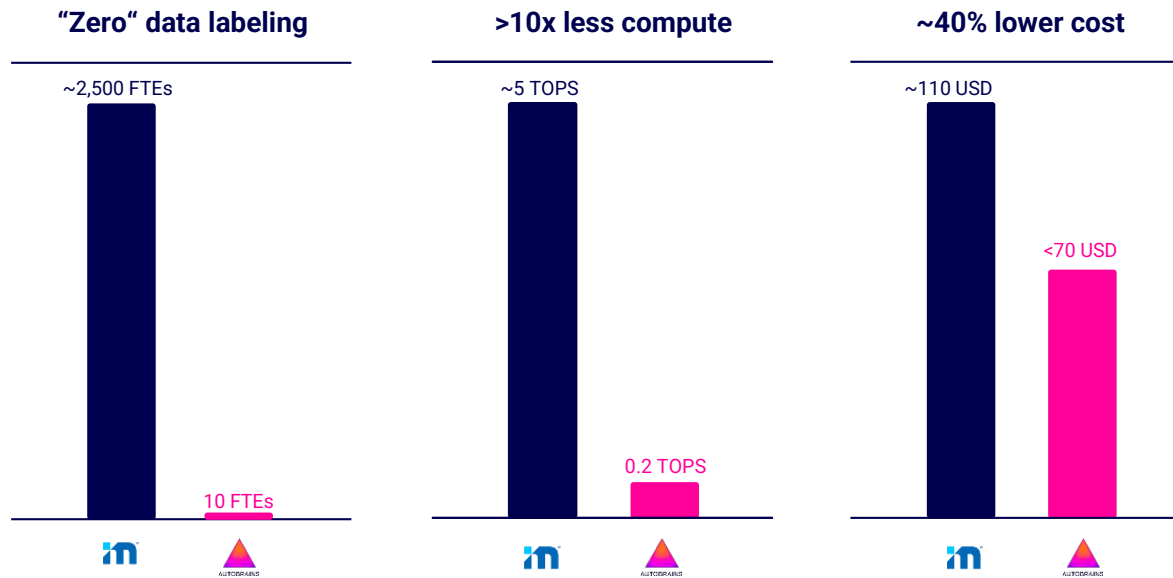
Product description

AB Entry L0-L2 	Cost efficient front camera solution, providing full regulatory compliance in low compute.	Up to 4 TOPS	1V 1.7MP 110°/8MP 120°	
AB Volume & Premium L2/L2+ 	- Complete L2+ perception stack with single SoC. - Processing front + parking cams + DMS.	Up to 10 TOPS	Up to 5V5R 2-8MP 120°	Sensor Fusion 
AB Luxury L3/L4 	Full edge case coverage based on break-through AI stack, including radar and lidar perception	Up to 100 TOPs	11V5R1L 360° perception	Sensor Fusion 

ADAS business case

Autobrain vs. Mobileye

Mobileye dominates market with >60% share and ~\$1.4bn annual revenue today



AB's Cartex perception solutions drastically outperform Mobileye's core solution, which generates more than \$1.1 billion in annual revenue today

Note: Mobileye EyeQ Series is core product generating approx. \$1.1bn annual revenue; Illustrative charts not to scale; figures derived from publicly available performance data and internal BOM-based reverse costing analysis; TOPS = Tera Operations per Second

Perception Fields

A novel approach to machine perception and autonomous driving

"Imagine trusting autonomous driving the same way we trust the laws of physics"

In the perception-based approach to autonomous driving, bounding boxes are replaced by perception fields:

- 1) A perception field is a virtual force field acting on the ego car
- 2) Ego motion is determined by the perception fields associated with all detected road objects
- 3) Decision making becomes trivial



Perception Fields closes the gap between perception and decision and provides a better:



Explainability



Generalizability



Robustness



Autobrain's Autolabelling – Autonomous Ground Truth

Automated labelling, unlimited by human annotation
Bulletproof solution for tagging of objects, in single and multi-frame.

- ✓ 2D+3D static/ dynamic labelling incl. distance, velocity, etc.
- ✓ Labels camera, radar and lidar recordings
- ✓ Added output, including dynamic labelling



Cost effective – No need for manual labelling

High quality - Accurate, consistent and quick

Sensor agnostic – tag any sensor input

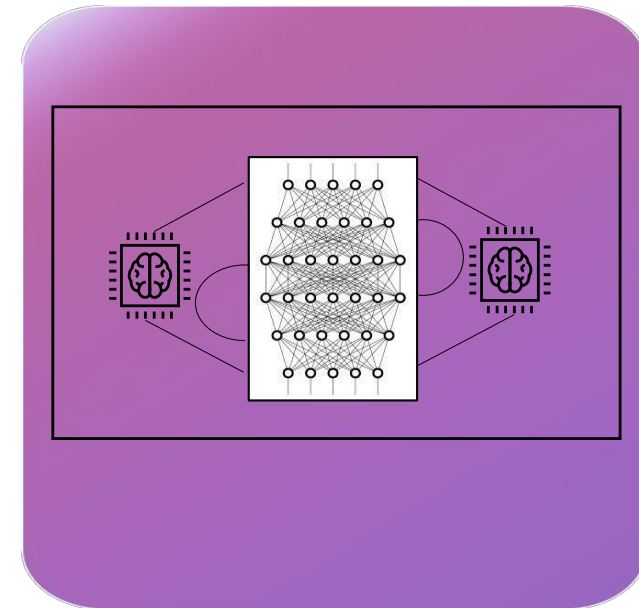
Scalable – Easily add more data classes and extended datasets

Autobrain Flow – Development platform for Real-World AI

Deep learning is facing "diminishing returns" problem - more data and larger networks cannot solve the tail end of edge cases(1).

Flow is a platform for quickly finding network issues, solving and easily improving the network.

1. **Assess network** - input deep learning network and automatically generate the set of unsolved edge cases
2. **Root problem** - Extend each edge-case to a benchmark by signature-based active learning
3. **Complete resolution** - Resolve the edge cases with signatures or by bypassing the neural net



Complementary solution – on top of existing network --
> No need to retrain the whole network

Scale any network to solve edge cases

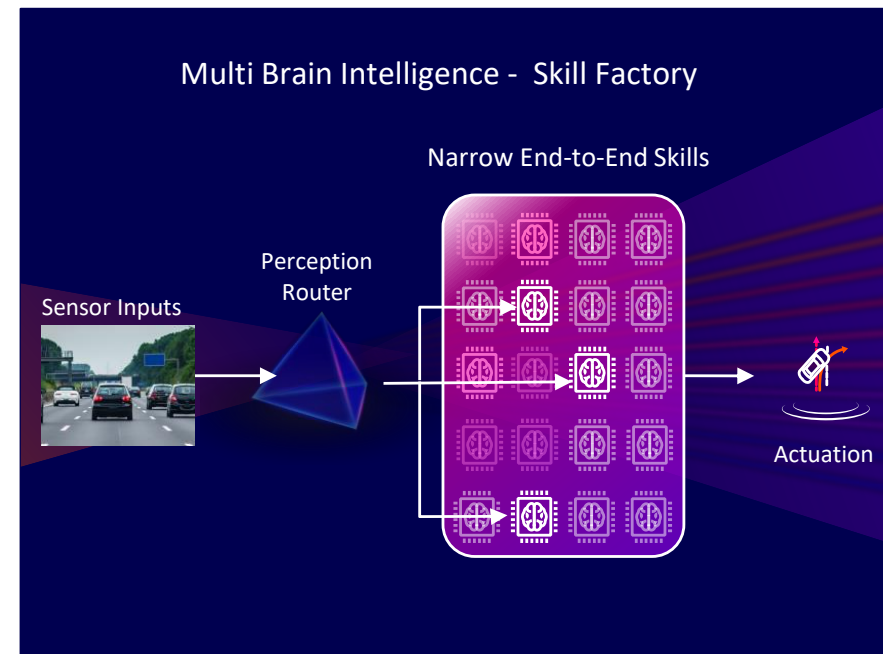
Accelerate debug time by using active learning to quickly penetrate gray/ black box DNN architectures

Autobrain Skills – Scaling Safely to L4

From generic AI to scalable AI

An ensemble of highly specialized narrow AI agents (“skills”) performing end-to-end tasks optimized for each, specific scenario

1. **Breaking the problem of driving:** Driving becomes a large, but finite set of problems
2. **Perception as a router:** The perception router is activating the right narrow agent/s based on the sensory input from the environment and the car’s mission/route
3. **Narrow skills:** Each skill perfectly solves narrow vertical end-to-end function/feature



Granular – licensing based on single skill

Accurate – focused on narrow scenario

Scalable – add more skills seamlessly

Explainable white box solution

Perception SW Stack Generations

Generation	Description	Issues	L1	L2	L3	L4	L5		
Gen 4	3D Surround Stack (Vision and Radar Fusion), self-learning							Self-Learning	
2022	=== Supervised Learning Wall ===								
Gen 3	3D Surround Stack (Vision, Lidar and Radar Fusion)	- Needs all scenarios labeled for training to mimic/repeat them = long tail - Highly sensitive to sensor type and location - Very high cost in training and execution (TOPS, Memory, Storage, Cost) - Strong connection between HW and SW						Supervised Learning	AI
Gen 2	2D Stack with individual DNNs (Lane, Free Space, Traffic Sign Recognition)	- Needs all scenarios labeled for training to mimic/repeat them = long tail - Highly sensitive to sensor type and location - Extreme high cost - Difficult to fuse together and conflict resolution							
2015	=== Computer Vision Wall ===								
Gen 1	Computer Vision Programmed								CV

AD Business Case

Only cost-efficient autonomous driving solution

through unique self-learning technology to achieve L3-L5 level of autonomy

AD Project *	Supervised Learning **	Self-/ Unsupervised Learning	Saving
SoC Cost	250 TOPS \$250	40% TOPS \$100	\$150 Million
Vehicle Program Adoptions	Sensor location and type changes \$38 Million	\$15 Million	\$17 Million
Labeling***	1000 Labeller \$80 Million	100 Labeller \$8 Million	\$72 Million
Total			\$239**** Million per project

Saving BOM Cost of **\$240!** per vehicle

* 4 Year AD L3+ Project with 1 Million Cars and 3 Car Programs (Medium, Premium, Luxury)

** 95 % of companies use supervised learning

*** Significant saving in labelling and training creates opportunity for auto-labelling solution

**** Additional savings through less training infrastructure (storage, SoC, memory), explainable solution (safety and risk) vs. black box, inferencing memory, higher level of autonomy not included yet in business case